

AY 2020-21

LIFE SKILLS (2020-21)	
Technical Paper writing	27-1-21 to 24-2-21
Patent Drafting	10-3-21 to 31-3-21



Bhavan's

(Founded in 1938 by Kulapati Dr. K. M. Munshi with the blessings of Mahatma Gandhi)

आ नो भद्राः क्रतवो यन्तु विश्वतः।

Let noble thoughts come to us from every side

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SARDAR PATEL INSTITUTE OF TECHNOLOGY

(An Empowered Autonomous Institute Affiliated to University of Mumbai)

Bhavan's Campus, Munshi Nagar, Andheri (West), Mumbai - 400 058, India

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The scheme for the Academic year 2020-2021 is as follows for Computer, Computer Science and Engineering-Data Science, Computer Science and Engineering-Artificial Intelligence, Electronics and Telecommunication and Master of Computer Applications.

Bhauddhraj

PRINCIPAL



2020-2024



Bharatiya Vidya Bhavan's

Sardar Patel Institute of Technology

(Autonomous Institute Affiliated to University of Mumbai)

[Knowledge is Nectar]

Liberal, Pi-Model of Engineering Education @ SPIT (Department of Computer Engineering)

CURRICULUM SCHEME FOR UNDERGRADUATE ACADEMIC PROGRAM (COMPUTER ENGG) AT SPIT

2020 ITERATION: COMPUTER DOMAIN (COMP Branch)

Salient Features

- 157-Credit **Liberal** Engineering Education Model.
- A strong **program core of 12 courses** and **6 baskets of program electives** to ensure the breadth and depth in a chosen domain of studies. Program electives are arranged either to grow in a specified vertical or have diversified exposure.
- **Full semester industry internship to interested students.**
- Aggressive model of “**Learning-by-doing**”. (Engagement in classroom and laboratory sessions is 50:50)
- Special tracks for “**Minor**” Certification for interested learners, ensuring significant awareness of additional discipline leading to multiple specializations
- **Unique, multi-track model of “Honors” Certification**, for well performers for enhanced depth in the domain of study.
- Special sequel of optional **industry floated “SCOPE”** courses (Skilled Certification for Outcome-based Professional Education) for interested learners, ensuring high technical skills, in the diversified cutting-edge technologies.
- **First-of-its-kind-in-education** blend to Engineering Curriculum. “**ABLL@LLC**”[®] (Activity Based Liberal Learning about **Life, Literature and Culture**) in all **EIGHT** semesters, ensuring **all dimensional holistic growth** of the learner. These eight activity based mini courses are offered as two sequels namely “**SEVA**”[®] (Social Empowerment through Various Activities), and “**SATVA**”[®] (Self accomplishment through various Activities).

This curriculum aims at development of an **all-rounded** personality. It follows **holistic** approach of education, ensures strong science, mathematics foundation and program core, develops expertise in domain vertical though sequel of electives, ensures significant exposure of additional discipline through “Minor” program, collaborates

outside world for the imparting relevant skills through “SCOPE” courses, challenges good learners through “Honors” evaluation, and systematically develops soft skills, and social, physical, mental, spiritual personality through carefully articulated **Liberal Learning** and **Humanities** sequels. Thus, it offers a unique, liberal “**Pi-Model**” of Engineering Education.

Program Core

At SPIT, every undergraduate program consists of **Twelve Core Courses** referred to as Program Core. Several academic models from reputed institutions in the country and outside the country are studied in articulating this Program Core, to make curriculum Globally Competitive. All courses in this Core have laboratory components to augment the learning. Each program core course has an additional optional component of “Contents beyond the curriculum” which is carefully designed to ensure additional 15-20 hours engagement of the learners. The learner thus is nurtured towards the “Self-Learning” and “lifelong learning” which are essential attributes of a 21st Century learner.

Program Electives

At SPIT, every program has **Six baskets** of Program Electives, each basket having a minimum 3 courses. This enables learner to grow in a **domain-specialization** or **domain-vertical**. For example, learner can graduate with B.Tech Electronics with a vertical in “Embedded Systems” or “VLSI” or “Signal Processing”. Or a learner can graduate with B.Tech Computer Engineering with specialization in “Security” or “ML & AI” or “Computer Networking” or “Data Science”. At the same time, learner can increase her bandwidth by opting for elective courses which are general in nature, not pointing out towards a specific vertical.

Open Electives

Every undergraduate program has three baskets of open electives. This is planned to give exposure to interdisciplinary and cross disciplinary domains. The courses in these baskets are planned both at department and institute level. Students can choose any combination of these courses (not floated by the parent department) to get familiar with other domains of learning. One of these open electives must be chosen from Basic science courses or Engineering Science courses. **This unique approach of offering additional basic science or engineering science elective at senior level aims at appreciating the importance of other domains of learning.**

Humanities and Social Science Electives

National Education policy 2019 has aptly spelled out the necessity of Humanities in the Professional Education. It quotes, “A holistic and liberal education as described so beautifully in India’s past is indeed what is needed for the education of India in the future to truly lead the country into the 21st century and the fourth industrial revolution. Even engineering schools such as the IITs must move towards a more liberal education integrating arts and humanities”. Every program at SPIT has three baskets of humanities. Learners are encouraged to take diversified courses in the field of languages, law, history, economics, management, finance etc.

SCOPE Certification

This unique sequel is designed to systematically develop skills required for an industrial sector. SPIT is partnering with various industries to offer the high-end skills required for a specific industrial sector. Well performing students can stretch the envelop and add new dimension to their Professional Personality by earning this certification. There are multiple tracks for SCOPE certification. Each track is offered with partnership with reputed institution or industry. These tracks are jointly designed by SPIT and partnering industry. Each track has four courses (modules). Each module/course is of 2-3 credits including laboratory component for most of the tracks. These tracks are also open for outside learners, leading to Certificate Program in a chosen domain.

Minor Certification

This additional and optional certification provides an opportunity to learner to develop the leaners in the additional domain of interests. It broadens the education and ensures the multi-disciplinary development which is essential attribute of 21st century engineer. However, this is optional. Well performing students can stretch the envelop and add new dimension to their Professional Personality. Each track for this minor certification is offered either by SPIT or with partnership with other reputed institutions. Each track has four courses (modules). Each course is of 3 credits and laboratory component if any. These tracks are also open for outside learners, leading to Certificate Program of 12 credits in a chosen domain.

Honors Certification

While the Minor and SCOPE certifications aim at adding additional professional dimension to the professional personality of the learners, the Honors certification gives opportunity to well performing learners to drive deep in the chosen field of study. Multiple plans/ways are planned to encourage learners to earn this certification which essentially excite the learners to push an envelope and go extra/deep in the chosen area of the study. Students earn additional stars (*) as shown in Table 1 during their program. If at the time of graduation student earns total **TWELVE** stars, she is conferred with “Honors” certification.

ABLL@LLC

- Students are required to earn 6 credits through 8 semesters.
- If student is not able attendance/performance requirements, he/she will be dropped from the course and will have to enroll in additional course in the next semester.
- A student can enroll in maximum 2 courses in a semester.

SEVA (Social Empowerment through Various Activities)			
Module	Title	Courses	CODE
SEVA-I	SOCHO BHARAT	Study of Green & White Revolutions in India	SV10
		Government Missions [Study of any 2]	SV11
		Study of India's top 2 problems	SV12
		Study of World's top 2 problems	SV13
		How Government Works? [Study of one department of the Central/ State Government]	SV14
		Study of one of the identified Books	SV15
		Study of two National policies	SV16
		Any other activity approved by Dean Academics	SV1X
Module	Title	Courses	CODE
SEVA-II	SWACCH BHARAT	River/Beach/Mohalla/School/Campus/Govt offices Cleaning	SV20
		Waste Segregation Surveys	SV21
		NSS camp in village for a week	SV22
		Medical camps in schools	SV23
		First Aid training for a week	SV24
		Surveys and Estimation for roof top solar	SV25
		NCC participation	SV26
		Any activity approved by Dean Academics	SV2X
Module	Title	Courses	CODE
SEVA-III	SHIKSHIT BHARAT	Mentoring of School Children	SV30
		Digital Literacy for yielders	SV31
		Value addition for deprived schools	SV32
		Mentoring junior (first year) students at SPIT	SV33
		Teaching Assistantship at SPIT	SV34
		Development of learning material for schools/ITIs	SV35
		Participation in "Teach-for-India" movement	SV36
		Any other activity approved by Dean Academics	SV3X
Module	Title	Courses	CODE
SEVA-IV	SAMRUDDHA BHARAT	Great Grass Root Innovations	SV40
		Innovation and Creativity	SV41
		Critical Thinking and Problem solving	SV42
		Team work and collaboration	SV43
		Leadership & Entrepreneurship	SV44
		Design Thinking	SV45
		Study of one of the identified books	SV47
		Work with START-UP at SPIT	SV48
		Any other activity approved by Dean Academics	SV49

Table 3: SATVA

SATVA (Self Accomplishment Through Various Activities)			
Module	Title	Courses	CODE
SATVA-I	SANSKRIT BHARAT	Values and Ethos of Bhavan	ST10
		Essence of Indian traditional knowledge	ST11
		Philosophy of religion (any)	ST12
		Study of Life Management / Kindle Life / Life Empowerment and Enriching Program or any other book cited.	ST13
		Study of any of GREAT sons of INDIA [Ex. Gandhi, Ambedkar, Phule, Savarkar, Sardar Patel, Nehru, Shivaji, JRD Tata etc]	ST14
		Any other course approved by Dean Academics	ST1X
SATVA-II	SAKSHAM BHARAT	Target based Physical Exercise for example-Running [Test 5 kms in a stretch], Swimming [Test 1 km in a stretch], Walking [Test 20 kms in a stretch], Trekking [7days], Cycling	ST20
		Sports – Representation of Institute at University level/Inter college level and above in ANY sport	ST21
		Participation in National Tech Fest, AICTE-Hackathon, Industry floated global and national competitions, Robocon, BAHA etc	ST22
		Yoga vidya -I	ST23
		Any other activity approved by Dean Academics	ST2X
		Institute representation in prestigious cultural fests/competitions	ST30
SATVA-III	SUNDER BHARAT	Dance [Bharatanatyam /Kathak /Lavani /Western Dance]. Only for beginners	ST31
		Learning musical instruments [Any type]. Only for beginners.	ST32
		Film Appreciation/Dramatics/Seeing through Painting	ST33
		Making short film/Photography	ST34
		Yogvidya-II	ST35
		Any other activity approved by Dean Academics and DOSA	ST3X
SATVA-IV	SURAKSHIT BHARAT	Food that Heals	ST40
		Personal and Social Hygiene	ST41
		Intellectual Property Rights	ST42
		Etiquette and Conversational skills	ST43
		Basics of Ayurveda	ST44
		Study of one of the identified Books	ST45
		Any other course approved by Dean Academics	ST4X

Sem II									
No	Type	Code	Course	L	T	P	O	E	C
1	BSC	MA102	Differential Equations and Complex Analysis	3	1	0	8	12	4
2	BSC	AS101	Engineering Physics	2	1	2	5	10	4
3	ESC	AS104	Engineering Graphics	1	0	4	2	07	3
4	ESC	ET101	Basic Electrical Engineering	3	0	2	6	11	4
5	ESC	CS102	Problem Solving using OOP	2	0	2	3	07	3
6	SBC	AS106	Skill Shop	0	0	2	0	02	1
7	ABL	SVIX/STIX	SEVA-I or SATVA-I	0	0	0	2	02	1
TOTAL				11	2	12	26	51	20

Sem III									
No	Type	Code	Course	L	T	P	O	E	C
1	BSC	MA203	Probability and Statistics	3	0	0	5	08	3
1	BSC*	MA202	Foundation of Mathematics-I*	2	1	0	6	09	3
2	PC	CS201	Discrete Structures and Graph Theory	3	0	0	4	07	3
3	PC	CS202	Data Structures	3	0	2	5	10	4
4	PC	CS203	Computer Architecture and Organization	3	0	2	4	09	4
5	PC	CS204	Database Management Systems	3	0	2	5	10	4
6	ABL	SVXX/STXX	SEVA II or III /SATVA II or III	0	0	0	3	03	1
7	HSSE	HSEX1	HSS-I	2	0	0	3	05	2
TOTAL				17	0	6	29	52	21

*Only for Lateral Entry Students

Sem IV									
No	Type	Code	Course	L	T	P	O	E	C
1	BSC	MA201	Linear Algebra	2	0	2	5	09	3
1	BSC*	MA204	Foundation of Mathematics-II	3	0	0	6	09	3
2	PC	CS205	Design and Analysis of Algorithms	3	0	2	5	10	4
3	PC	CS206	Operating Systems	3	0	2	5	10	4
4	PC	CS207	Computer Communications and Networks	3	0	2	5	10	4
5	SBC	CS208	Mini Project-I	0	0	0	4	04	2
6	ABL	SVXX/STXX	SEVA II or III /SATVA II or III	0	0	0	3	01	1
7	HSSE	HSEX2	HSS-II	2	0	0	3	05	2
8		AS201	Professional Communication Skills	1	0	2	2	05	2
9	S/M	SCX1/MNX1	SCOPE-I/Minor-I						3
TOTAL				14	0	10	32	54	22

Sem V									
No	Type	Code	Course	L	T	P	O	E	C
1	PC	CS301	Theory of Computation	3	0	0	6	9	3
2	PC	CS302	Software Engineering	3	0	2	5	10	4
3	PC	CS303A/ CS303B	Cryptography and System Security/Artificial Intelligence and Machine Learning	3	0	2	5	10	4
4	PC	CS304	Distributed Computing	3	0	2	5	10	4
5	SBC	CS305	Internet Technology Lab	1	0	2	5	08	2
6	ABL	SVXX/STXX	SEVA II or III /SATVA II or III	0	0	0	2	02	1
7	HSSE	HSEX3	HSS-III	2	0	0	3	05	2
8	S/M	SCX2/MNX2	SCOPE-II/Minor-II						3
TOTAL				15	0	8	31	54	20

Sem VI (Cat 1- For Students who have NOT preferred semester long internship)									
No	Type	Code	Course	L	T	P	O	E	C
1	OE	OEXXX	Open Elective-I	2	0	2	4	8	3
2	PC	CS306	Compiler Construction	3	0	2	5	10	4
3	PC	CS307	Foundation of Signal Processing	3	0	2	5	10	4
4	PE	CS3X1	PE-I	2	0	2	4	8	3
5	PE	CS3X2	PE-II	2	0	2	4	8	3
6	SBC	CS308	Mini Project-II	0	0	0	8	08	3
7	ABL	SVXX/STXX	SEVA II or III /SATVA II or III	0	0	0	3	03	1
8	S/M	SCX3/MNX3	SCOPE-III/Minor-III						3
TOTAL				12	0	10	33	55	21

Sem VII									
No	Type	Code	Course	L	T	P	O	E	C
1	OE	OEXXX	OE-II	2	0	2	4	8	3
2	OE	OEXXX	OE-III*	2	0	2	4	8	3
3	PE	CS4X3	PE-III	2	0	2	4	8	3
4	PE	CS4X4	PE-IV	2	0	2	4	8	3
5	SBC	CS401	Main Project Stage-I / Mini-project	0	0	0	4	4	3
6	ABL	SVXX/STXX	SEVA-IV/SATVA-IV	0	0	0	4	4	1
7	S/M/H	SCX4/MNX4/HOXX	SCOPE-IV/Minor-IV/Honors-I						3
TOTAL				8	0	8	24	40	16
*OE-III must be from Basic Science Elective or Engineering Science Elective									

Sem VIII (Option A: Cat1/Cat2)									
No	Type	Code	Course	L	T	P	O	E	C
1	OE *	OEHXX	OE-IV	2	0	2	4	8	3
2	PE	CS4X5	PE-V	2	0	2	4	8	3
3	PE	CS4X6	PE-VI	2	0	2	4	8	3
4	SBC	CS402	Main Project Stage-II	0	0	0	12	12	6
5	ABL	SVXX/STXX	SEVA-IV/SATVA-IV	0	0	0	4	04	1
6	H	HOXX	Honors-II						3
*May be taken from MOOCs, Essentially Humanities, Management related									
TOTAL				6	0	6	28	40	16

Sem VIII (Option B: Only for Cat1 students)									
No	Type	Code	Course	L	T	P	O	E	C
2	SBC	CS403	Industry Internship / Major Project	0	0	0	36	36	15
3	ABL	SVXX/STXX	SEVA-IV/SATVA-IV	0	0	0	4	04	1
4	H	HOXX	Honors-II						3
*May be taken from MOOCs, Essentially Humanities, Management related									
TOTAL				0	0	0	40	40	16



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LIFE SKILLS

patent writing

By

Sardar patel institute of technology

From: 10/03/21
to 31/03/21



Plisht

Dead Academics





Bharatiya Vidya Bhavan's
Sardar Patel Institute of Technology

Bhavan's Campus, Munshi Nagar, Andheri (West), Mumbai-400058-India
(Autonomous College Affiliated to University of Mumbai)

TECHNICAL PAPER WRITING

**DATE: 27-1-21 to
24-2-21**

A handwritten signature in blue ink, appearing to read 'Dead Academics', with a horizontal line underneath.

Dead Academics

